Part 1: Implement the second-price sealed-bid auction

Take two bidders — bidder 1 and bidder 2. Bidder 1's value v can be from 3 possible values: {0.2, 0.5 and 0.8}. For a given experiment, bidder 1's value will be fixed to one of the 3 values. Bidder 2's is the rival bidder and its value is drawn independently from a uniform distribution [0, 1]. The rival bidder(s) can adopt 3 possible strategies: (a) bid truthfully, (b) bid half of its true value, (c) bid a randomly drawn number from the uniform distribution [0, 1] and ignore its true value.

Bidder 1 bids truthfully. It pays the second highest bid if it wins and thus makes a profit (utility) of v-b, where v is its value and b is the rival bid. If bidder 1 loses, the profit (utility) it 0.

A. For each of the bidder 1's value {0.2, 0.5 and 0.8} and each rival strategy, try every bid for bidder 1 b=0.00, 0.01, …1.00 (this makes the horizontal axis). For each bid of bidder 1, simulate at least 100000 auctions where the bidder 2 draws its value randomly and bids according to its strategy. Plot the average utility of bidder 1 for each of its bid amount. There will be 9 plots — for each combination of bidder 1's fixed true value and the strategy adopted by the rival. Do these plots help bidder 1 to select its strategy?

B. Verify that truthful bidding by bidder 1 will be effective against every bid adopted by bidder 2. Fix the rival bid at specific values {0.1, 0.4, 0.9}. Confirm that b=v for bidder 1 gives the highest utility in each case.

C. Repeat parts A and C with two rival bidders — bidder 2 and bidder 3. Assume that both bidder 2 and bidder 3 will adopt the same strategy and bidder 3 will also draw its true value from a uniform distribution [0, 1].

Part 2: Implement the first price auction for n bidders. Bidder 1 deviates from the equilibrium and bids k.v where v is the true value and fraction k is selected from a grid (0.1 … 2.0). The remaining n-1 bidders stick to their equilibrium bid $((n-1)/n)\cdot v$. Estimate the expected utility for bidder 1 for each possible bid (k.v). Other bidders draw valuations from a uniform distribution. Measure the fraction of auctions for which the winner could have paid less and still won. Show for n=2, 3, 4.

Part 3: Implement VCG mechanism for buying the cheapest path from A to F where each edge is an agent with its private cost. Use Figure 7.3 of the book with the costs as given and reproduce the chosen path ABEF and the payments in the text: AB is paid 4, BE is paid 2, EF is paid 3, every other edge 0. Pick one edge on the path and one off it. For each, try every misreported cost from 0 to 10 in steps of 0.5 and examine the change in utility.